3D visualization and processing with Napari

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CZI Imaging Scientist

search Ctrl K

napari: a fast, interactive viewer for multi-dimensional images in Python

search Ctrl K

- Getting started
- napari tutorials
  - Annotation
  - Processing
  - Segmentation
    - Annotating segmentation with text and bounding boxes
  - Tracking
- How-to guides
- In-depth explanations
- Glossary
- napari workshops
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Annotating segmentation with text and bounding boxes

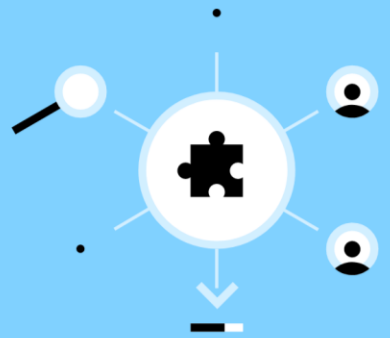
In this tutorial, we will use napari to view and annotate a segmentation with bounding boxes and text labels. Here we perform a segmentation by setting an intensity threshold with Otsu's method, but this same approach could also be used to visualize the results of other image processing algorithms such as object detection with neural networks.

ON THIS PAGE

napari **hub**

Plugins

Discover, install, and share napari plugins



- Discover plugins that solve your image analysis challenges
- Learn how to install into napari

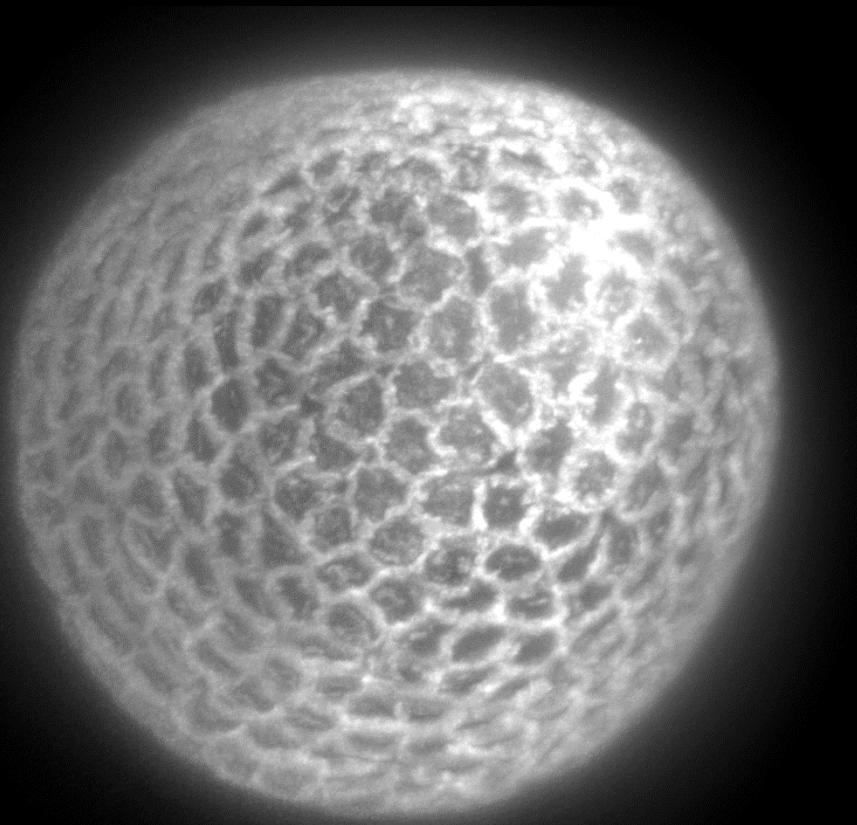
Share your image analysis tools with napari's growing community

Discover Plugins Browse all

Search for a plugin by keyword or author

In an **Anaconda prompt**:

- conda create -n NapariEnv python=3.10 openjdk=8 // create a virtual environment called NapariEnv
- conda activate NapariEnv // activate the virtual environment NapariEnv
- pip install napari[all] // install Napari
- napari // open Napari



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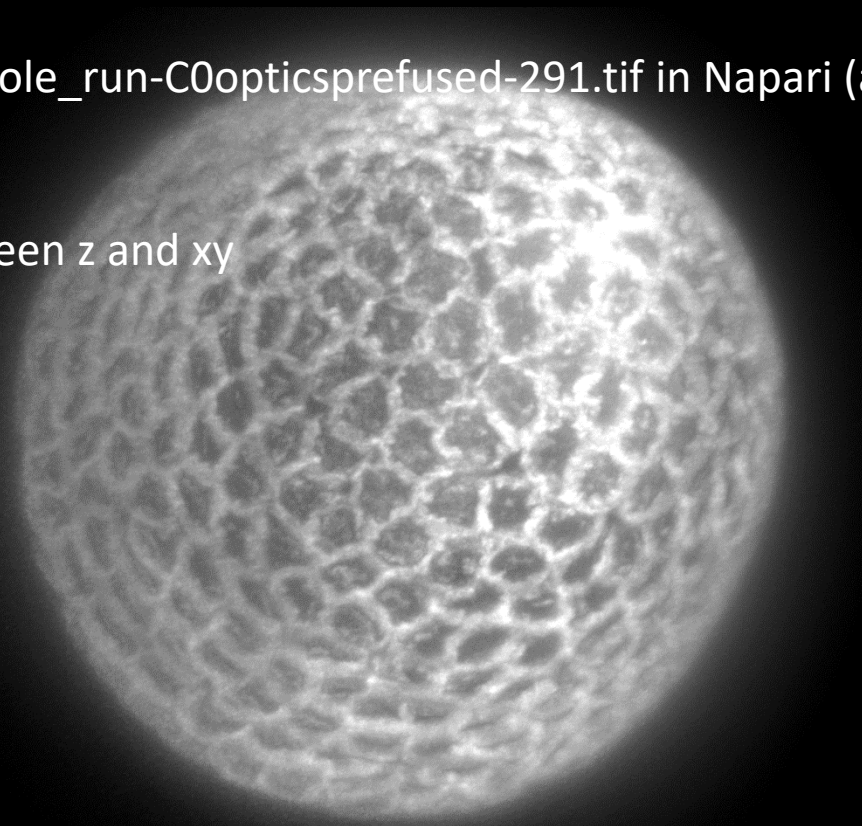
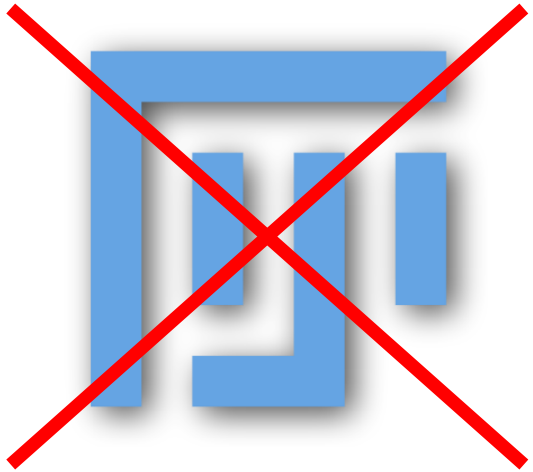
- Drag and drop Strausberg_Tribolium_LA-GFP_tailpole_run-C0opticsprefused-291.tif in Napari (available at https://zenodo.org/records/3981193)
- Visualize image in 2D and 3D, change scaling between z and xy
- Change contrast, change colormap, blending, ...
- Add Points layer
- Add shape layer
- Add a label layer, annotate 2 cells



Visualization

Visualization



- The **whole image** is loaded into the computer's **RAM** memory when opened with Fiji
- Stacks acquired with **lightsheet** microscopes = tens of GB

Visualization

- Open as virtual stack
- Save and visualize it with BigDataViewer

Visualization

- Open as virtual stack
- Save and visualize it with BigDataViewer

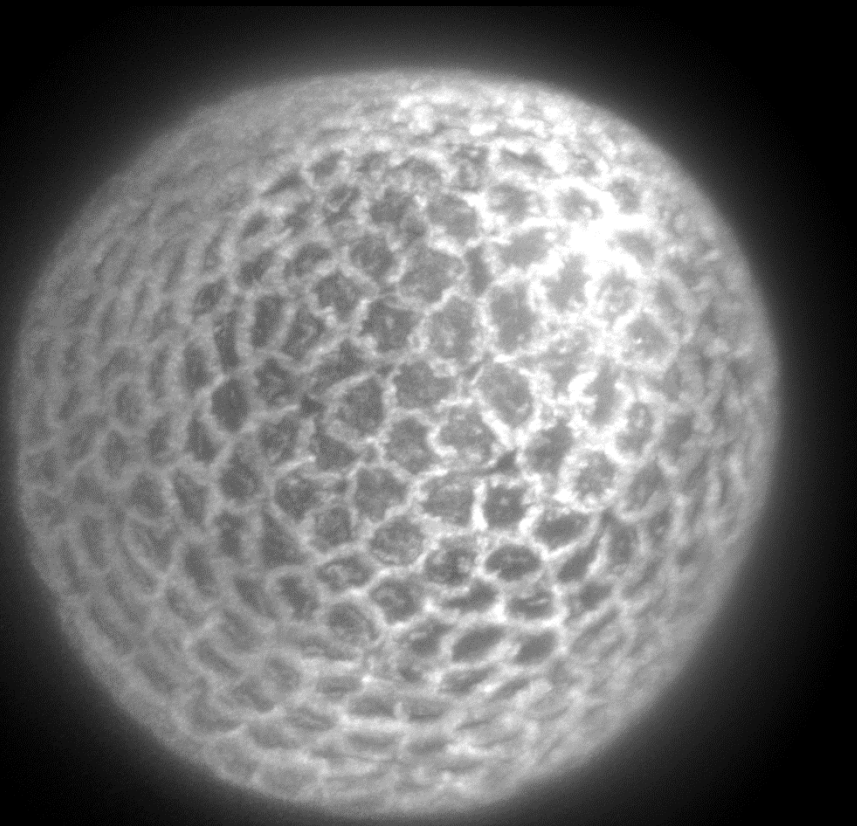
Zarr

napari

- Zarr format **splits data into chunks** so only **one or several chunks** are loaded **at once** into the computer's **RAM** memory
- **Napari**, a Python visualizer, allows to **visualize zarr data**

In a Miniforge **prompt**:

- conda install -c ome bioformats2raw // install Python package bioformats2raw
- bioformats2raw Strausberg_Tribolium_LA-GFP_tailpole_run-C0opticsprefused-291.tif Strausberg_Tribolium_LA-GFP_tailpole_run-C0opticsprefused-291.zarr // convert to zarr
- pip install napari-ome-zarr // install Napari plugin to open zarr images



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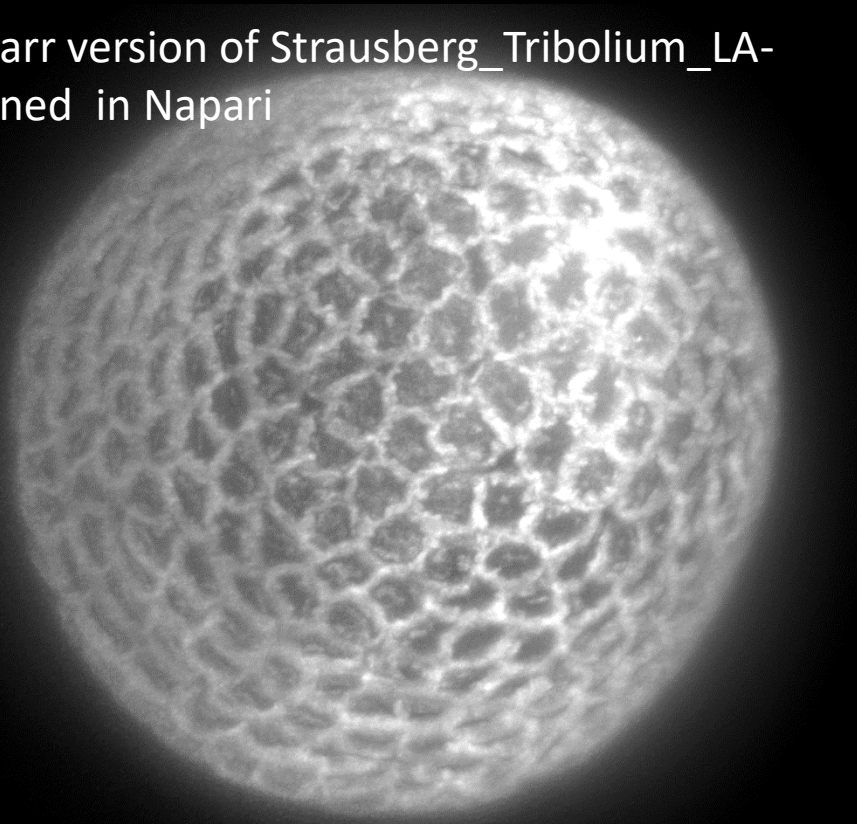
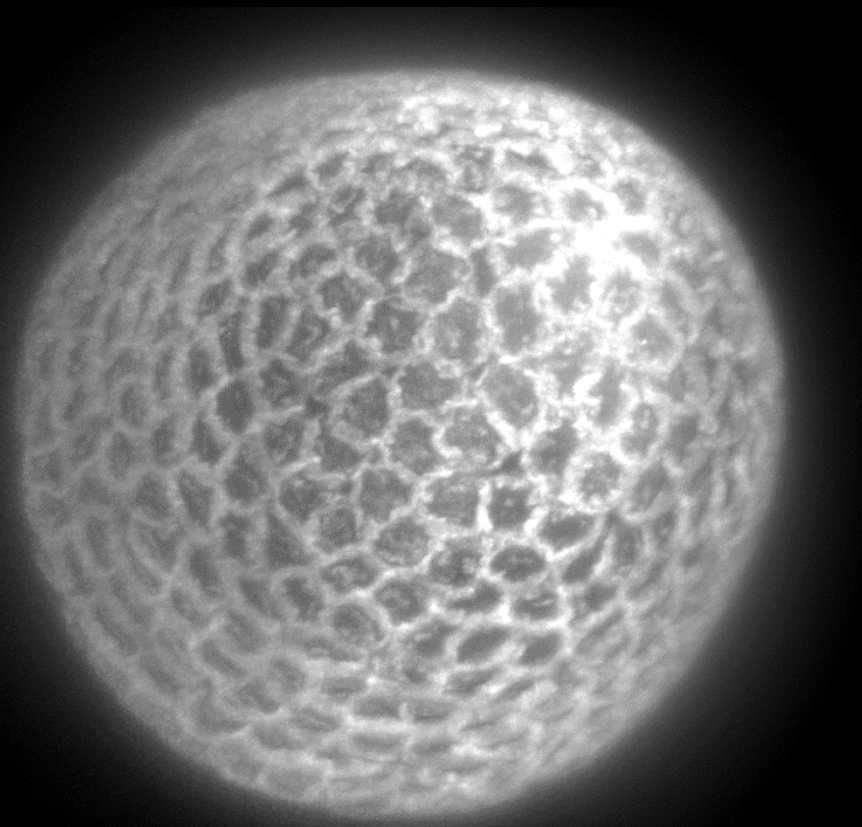
- bioformats2raw Strausberg_Tribolium_LA-GFP_tailpole_run-C0opticsprefused-291.tif Strausberg_Tribolium_LA-GFP_tailpole_run-C0opticsprefused-291.zarr // convert to zarr
- pip install napari-ome-zarr // install Napari plugin to open zarr images

- Compare the RAM usage between the tif and the zarr version of Strausberg_Tribolium_LA-GFP_tailpole_run-C0opticsprefused-291 when opened in Napari



In a Miniforge **prompt**:

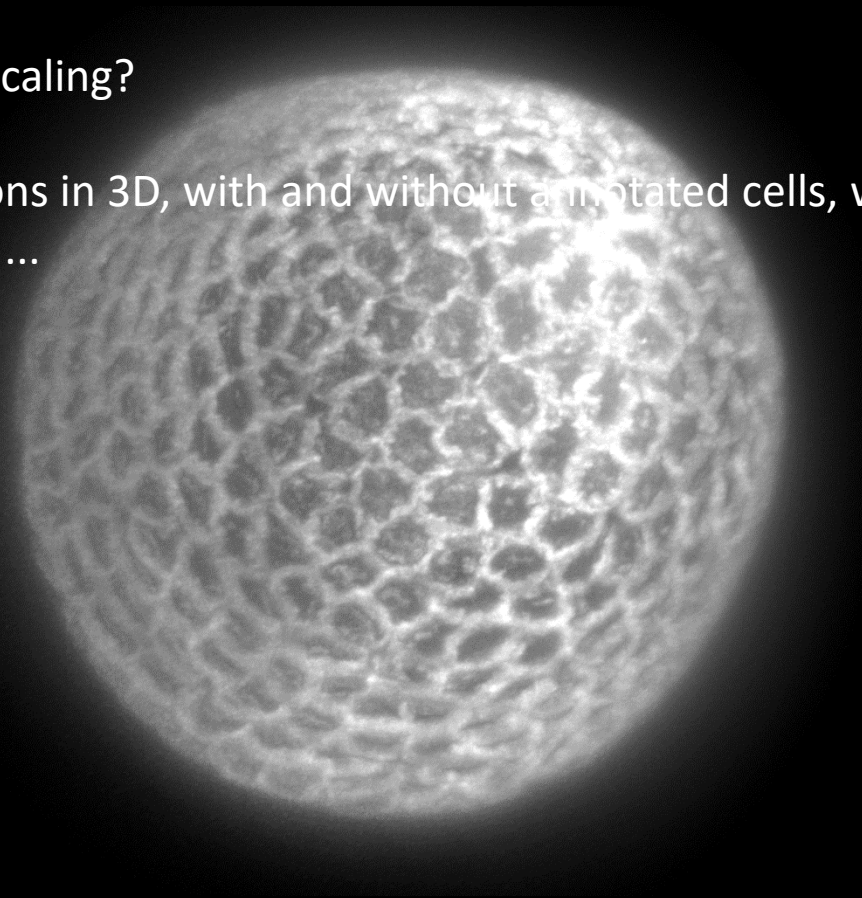
- pip install napari-animation // install Napari plugin to create videos



In a Miniforge **prompt**:

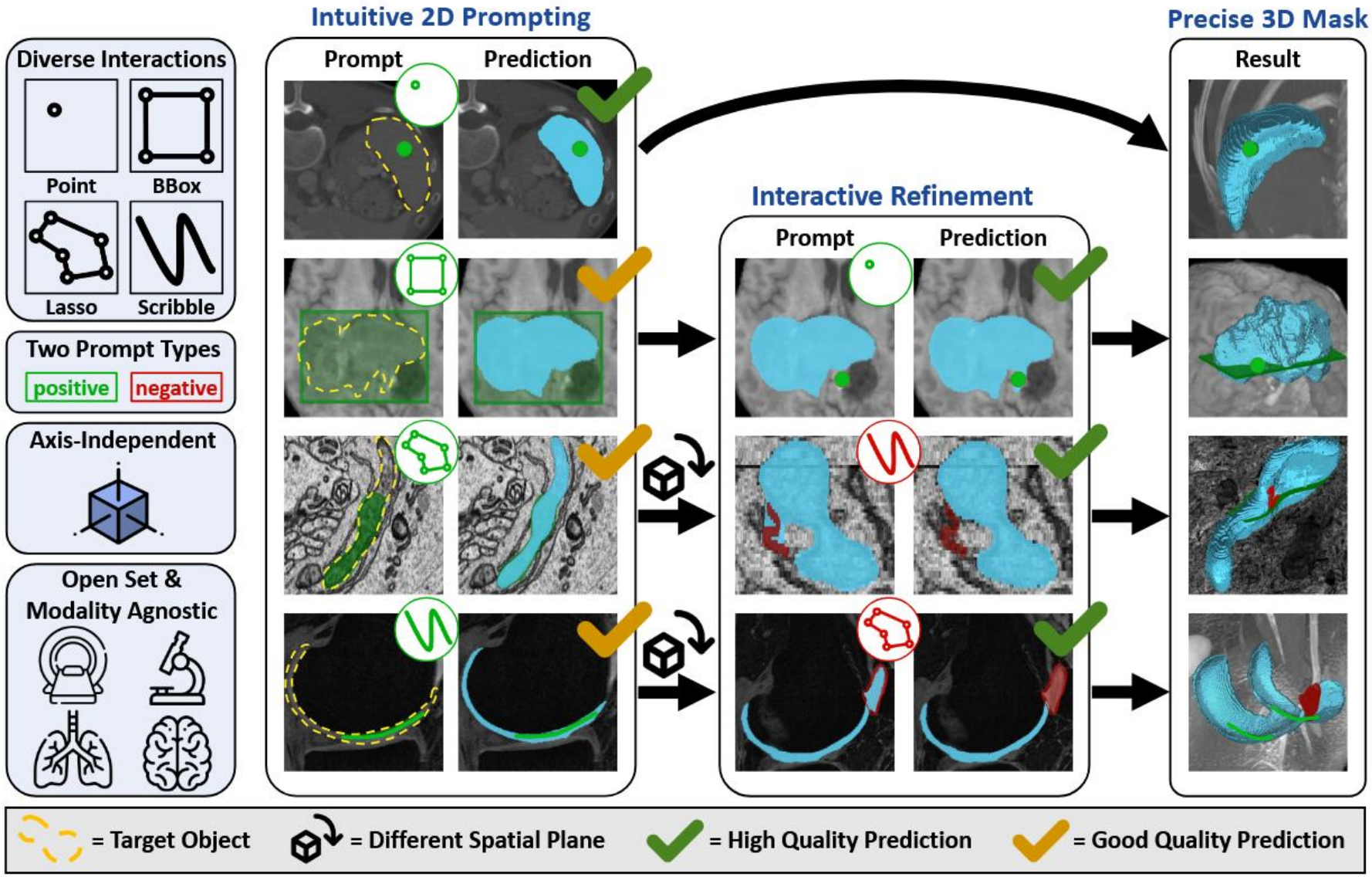
- pip install napari-animation // install Napari plugin to create videos

- Visualize image reconstruction in 3D, what about scaling?
- Create an animation with sections in 2D and sections in 3D, with and without annotated cells, with and without image, zooming in, zooming out, rotating, ...



nnInteractive: Redefining 3D Promptable Segmentation

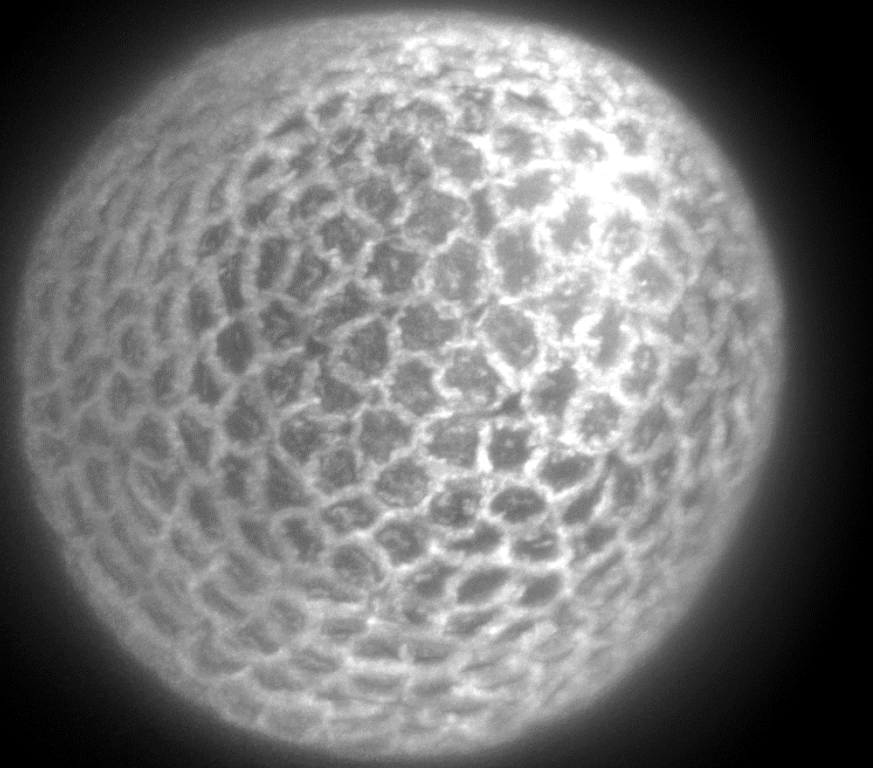
Fabian Isensee[1,4*], Maximilian Rokuss[1,2*], Lars Krämer[1,4*], Stefan Dinkelacker[1], Ashis Ravindran[1], Florian Stritzke[5], Benjamin Hamm[1,3], Tassilo Wald[1,2,4], Moritz Langenberg[1,2], Constantin Ulrich[1], Jonathan Deissler[1,2], Ralf Floca[1], Klaus Maier-Hein[1,2,3,4,6,7]



In a Miniforge **prompt**:

- pip install napari-nninteractive // install nnInterative

- Use nnInteractive to interactively segment cells, play with different views for prompts
- Create an animation with sections in 2D and sections in 3D, with and without annotated cells, with and without image, zooming in, zooming out, rotating, ...



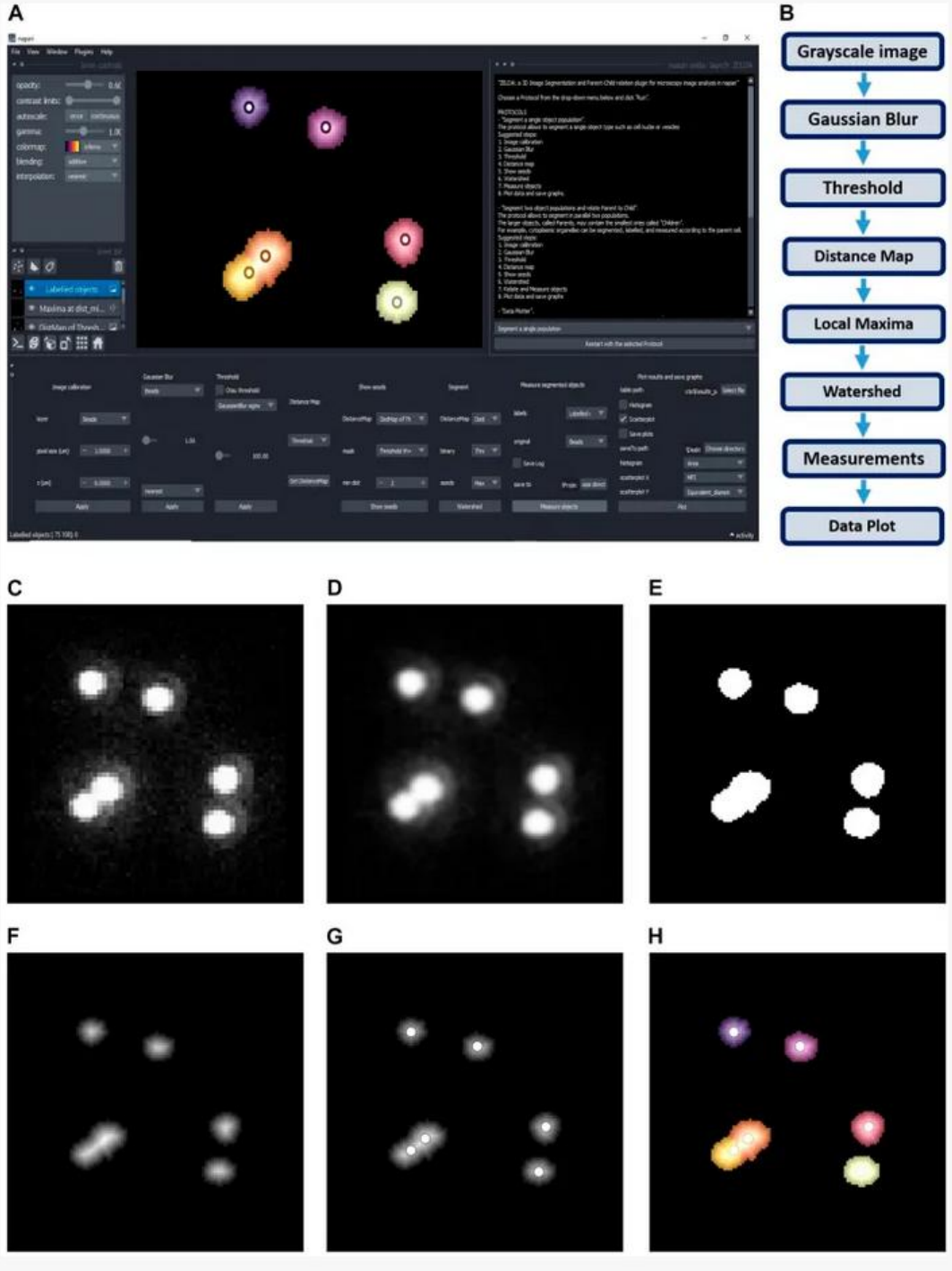
BRIEF RESEARCH REPORT article

Front. Comput. Sci., 04 January 2022

Sec. Computer Vision

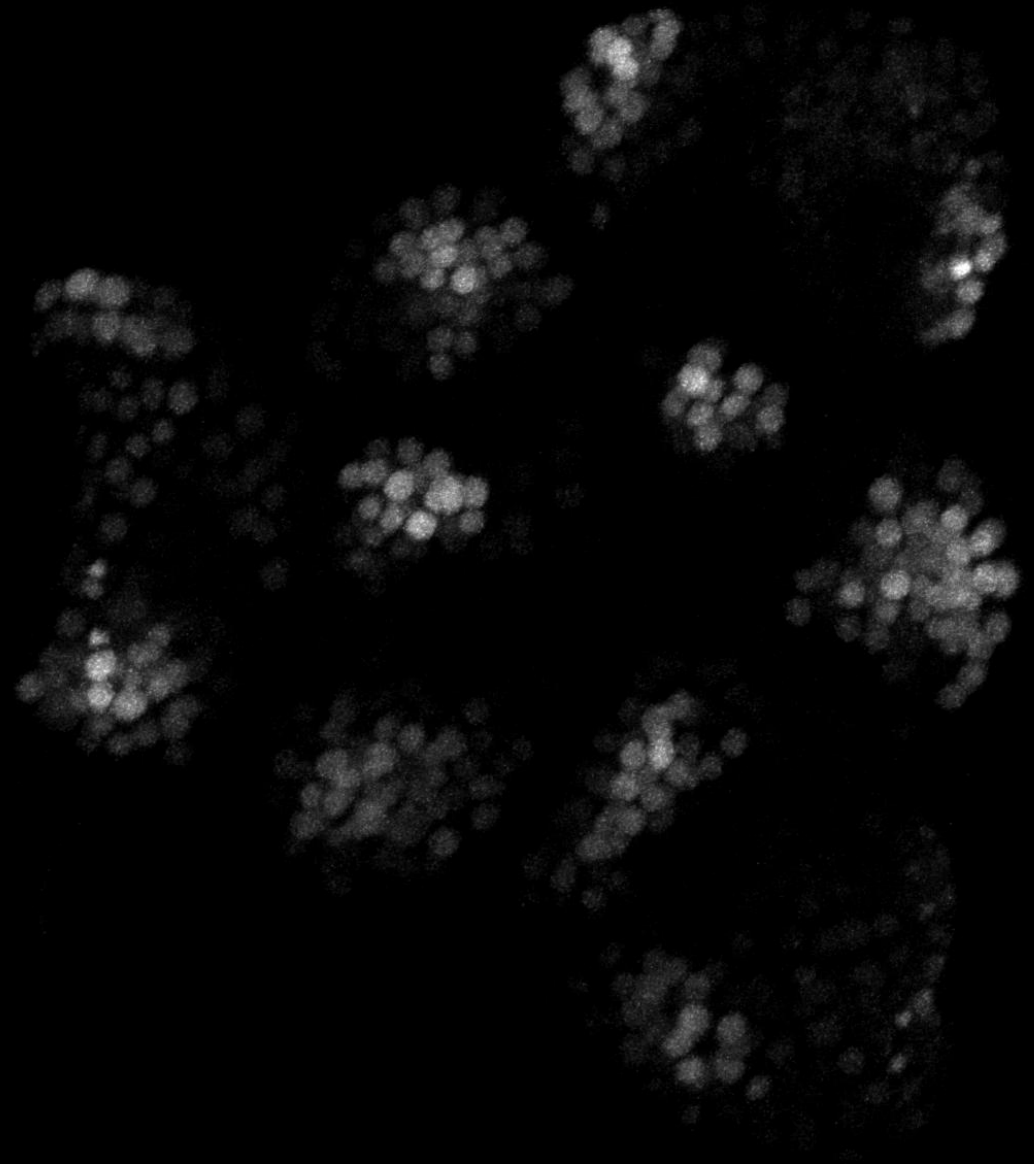
Volume 3 - 2021 | https://doi.org/10.3389/fcomp.2021.796117

ZELDA: A 3D Image Segmentation and Parent-Child Relation Plugin for Microscopy Image Analysis in *napari*



In a Miniforge **prompt**:

- pip install napari-zelda

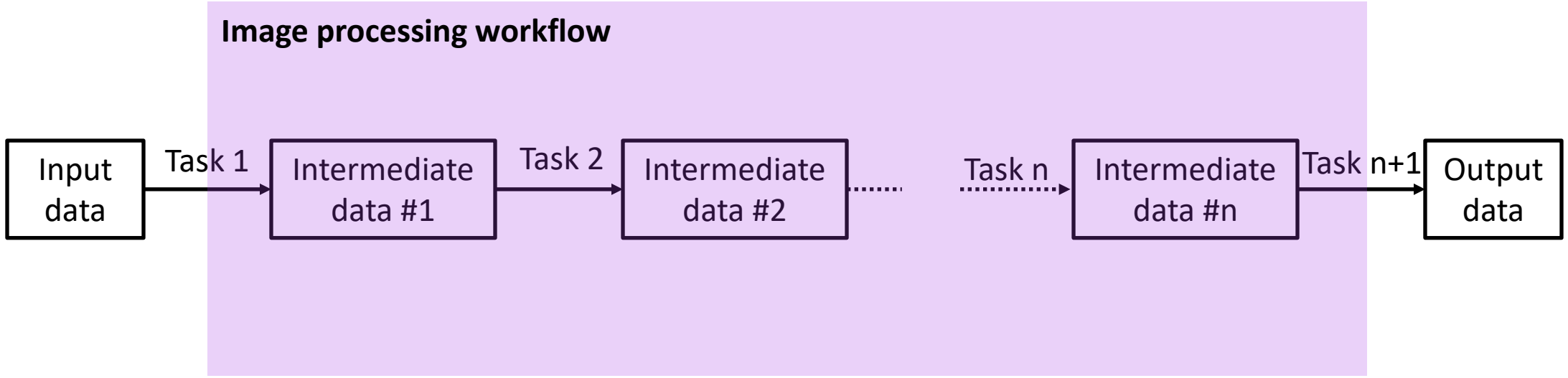


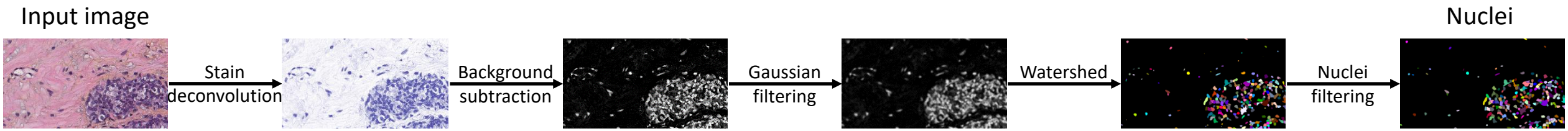
In a Miniforge **prompt**:

- pip install napari-zelda

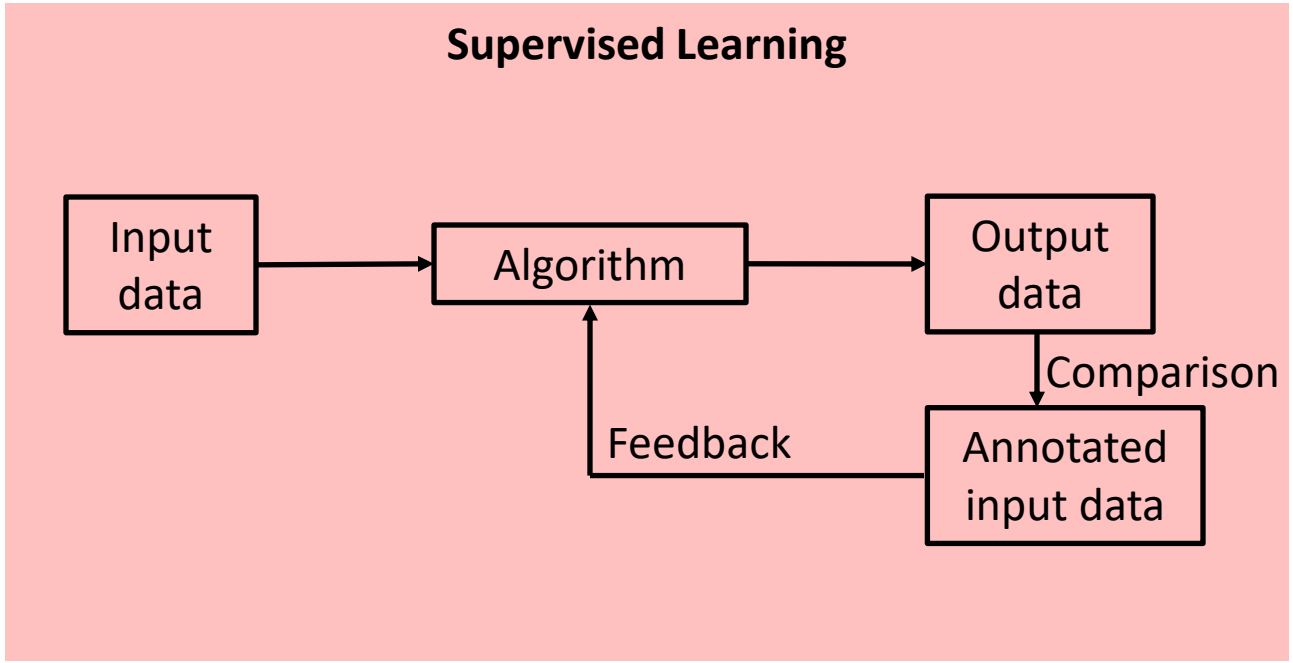
- Open MutX_DR5v2_6_4_M1_nuclei.tif (available at https://osf.io/uzq3w/)
- Use the protocol *Segment a single population* to segment nuclei with Napari-Zelda

EXPLICIT PROGRAMMING

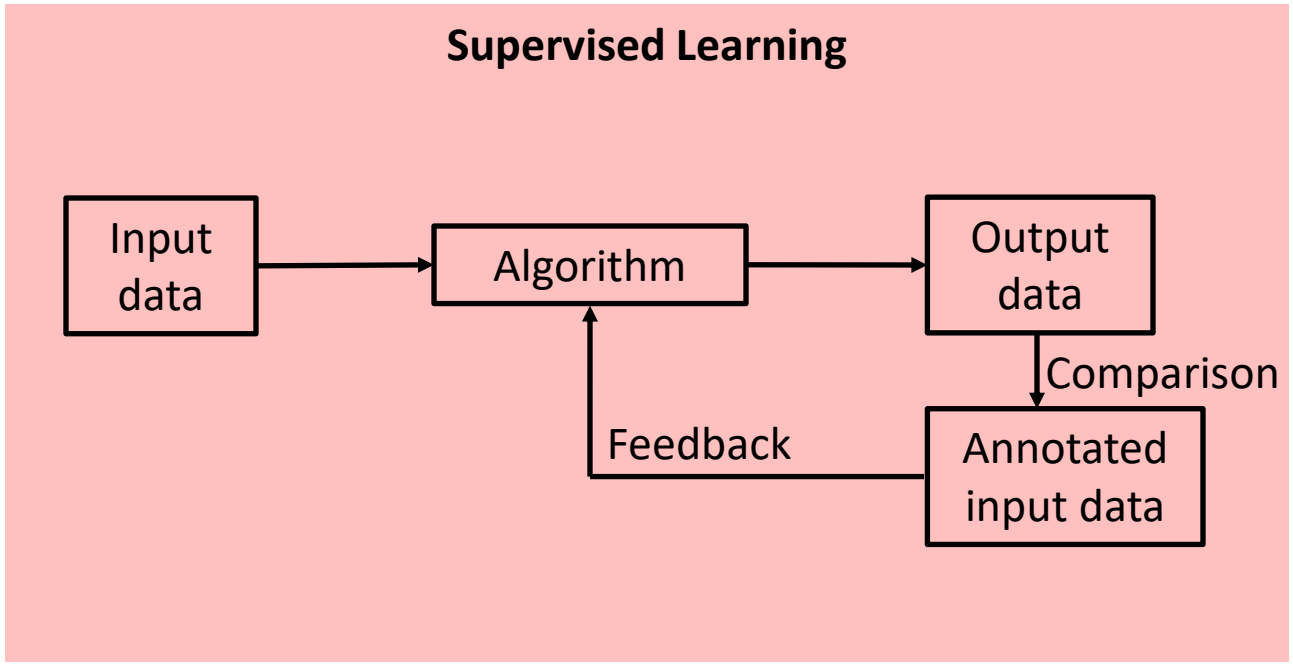


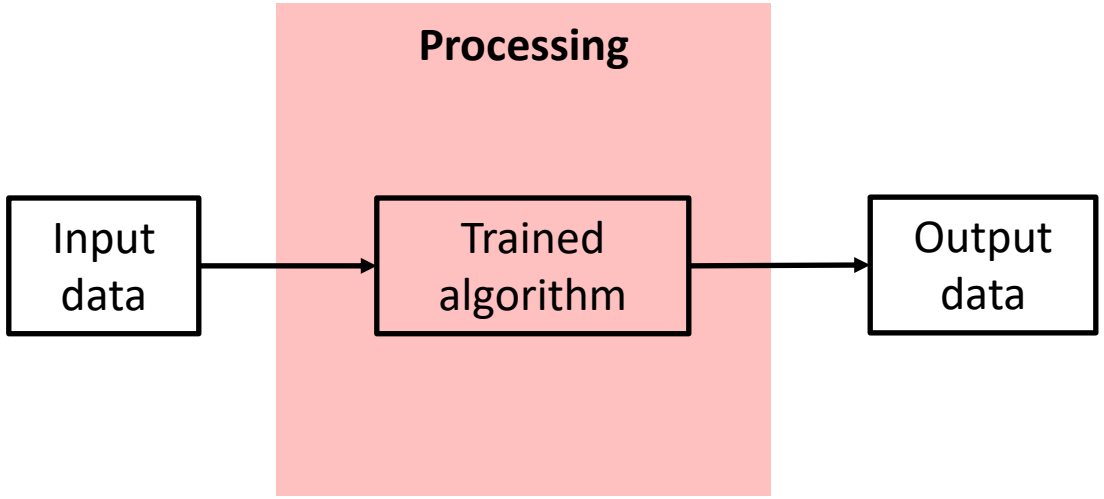


Supervised Machine Learning



Supervised Machine Learning





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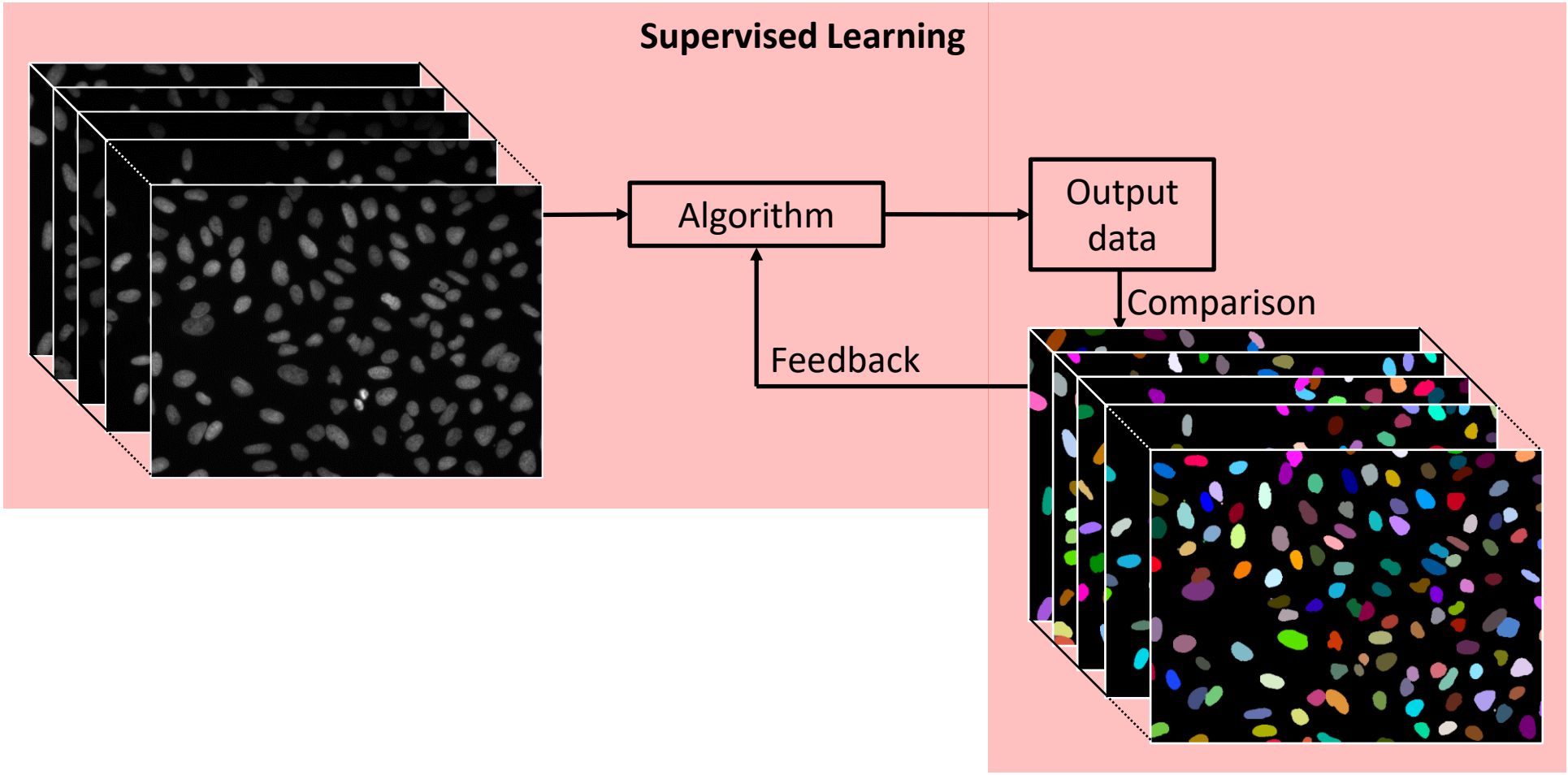
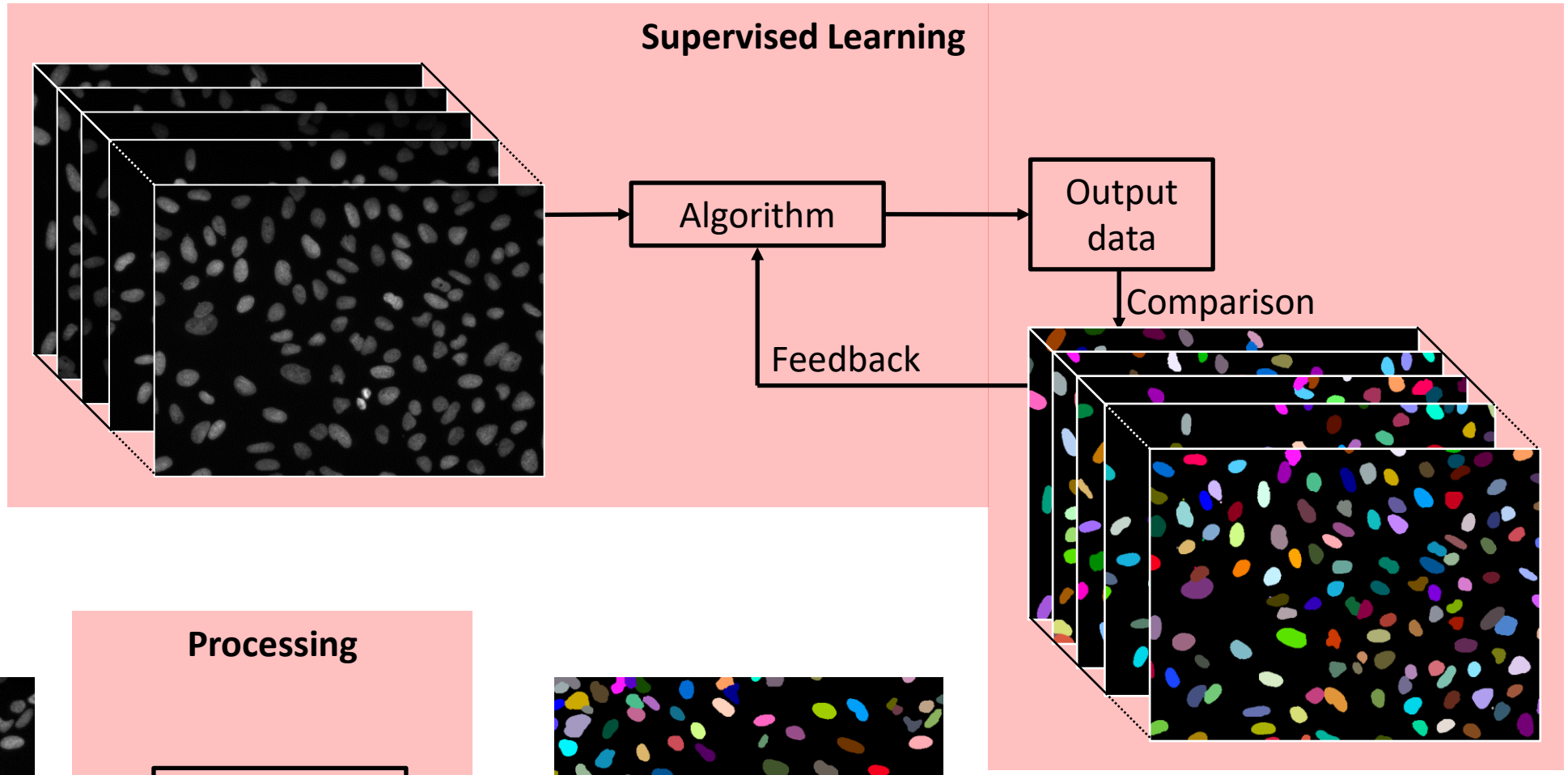
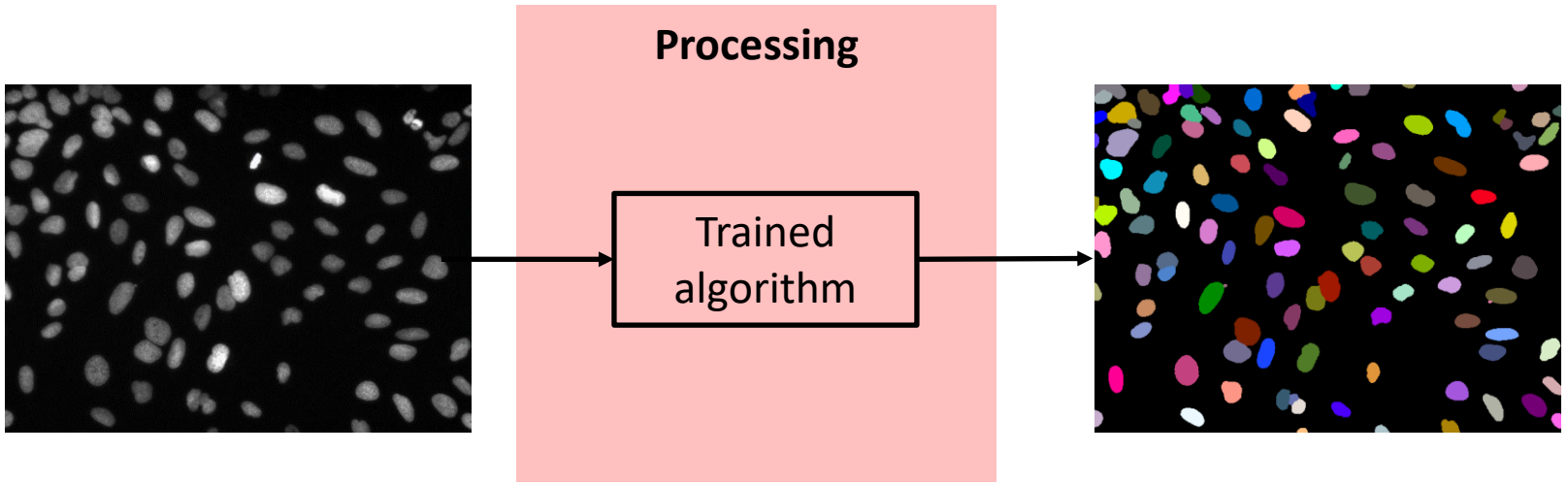


Supervised Machine Learning



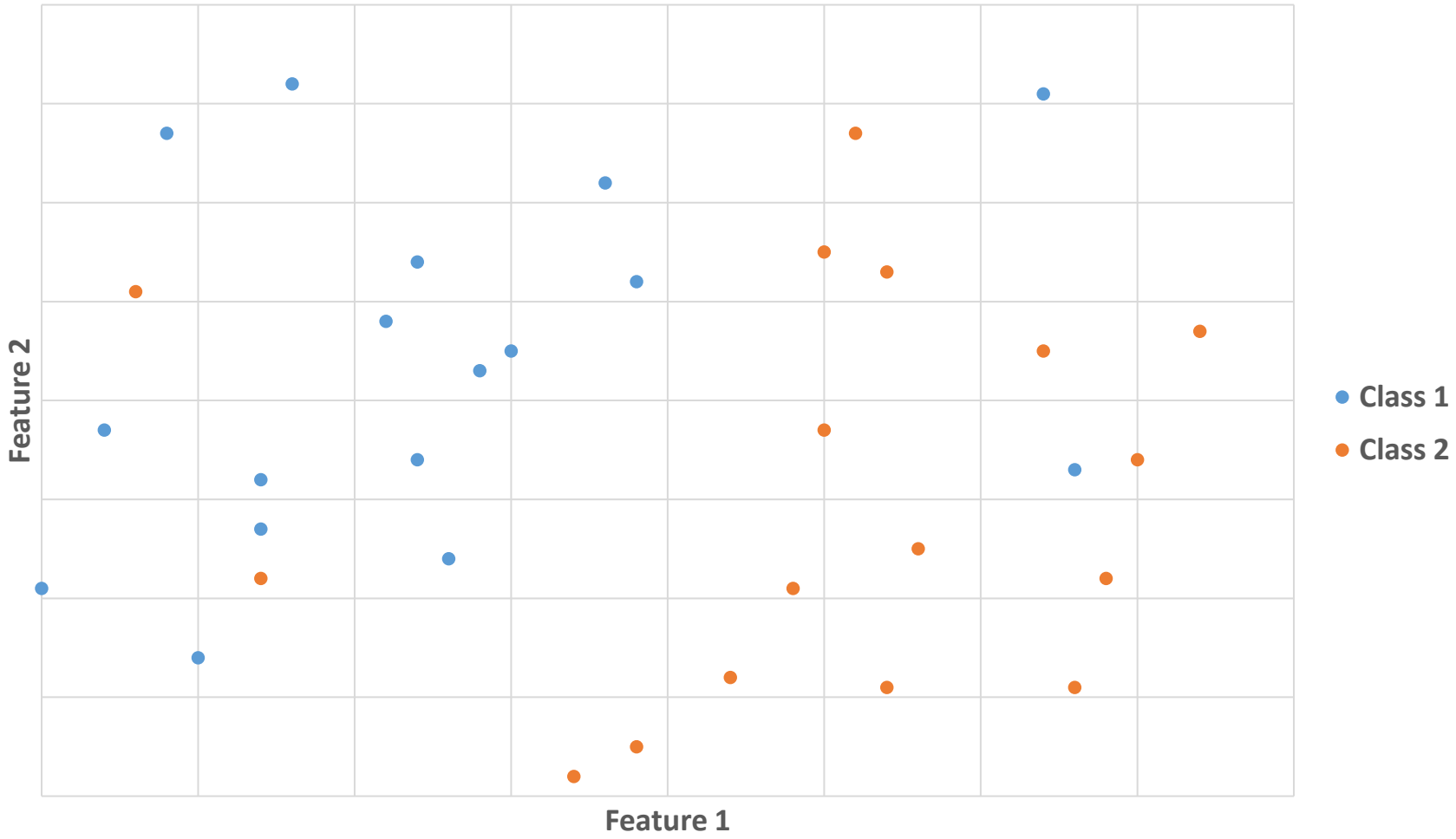


Shallow Machine Learning for Pixel Classification

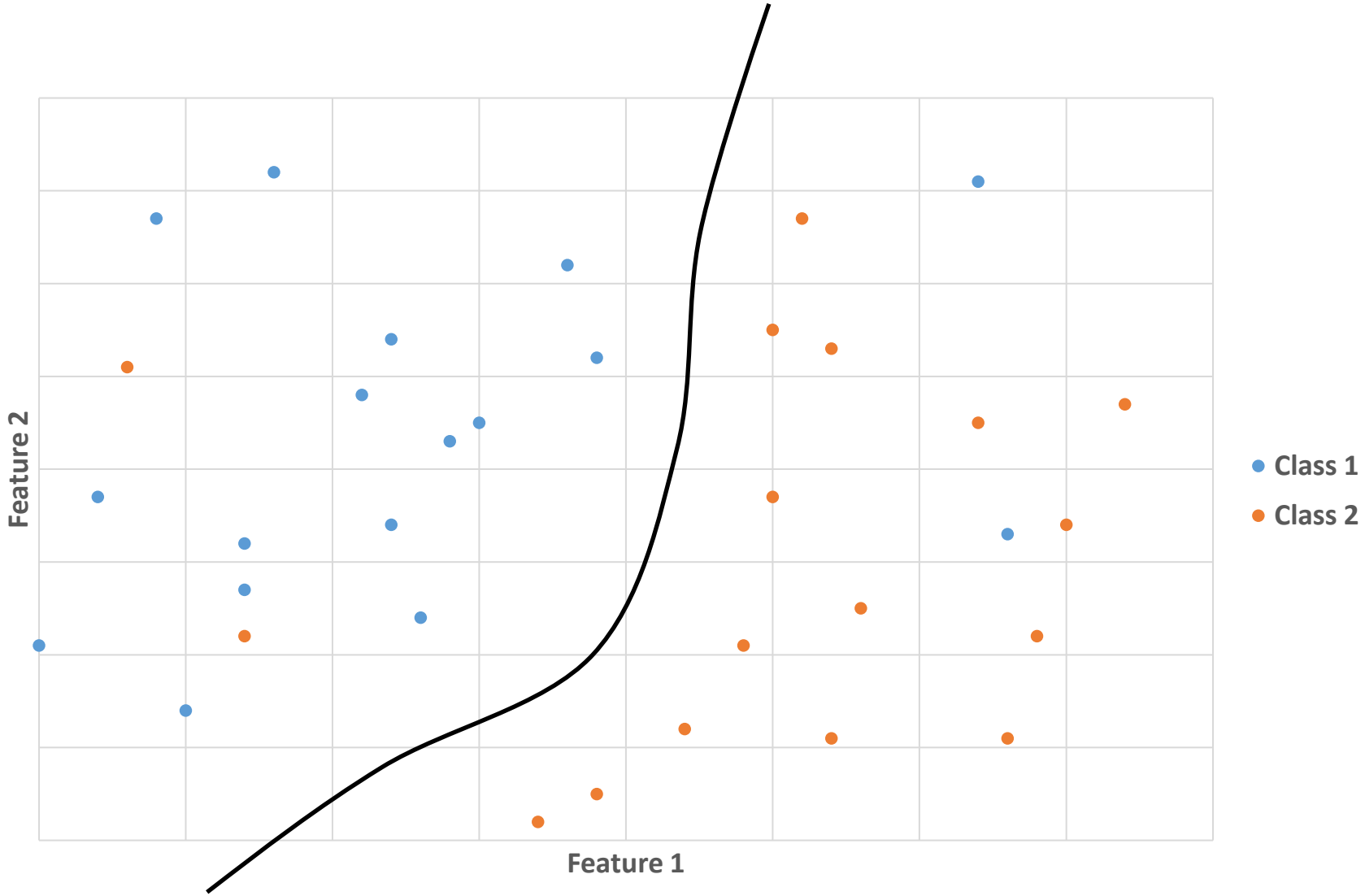
Supervised classification:

- **Examples** of classes are **manually** defined by the user
- A **classifier** is **trained** by using **defined features** with these examples
- Data is then **automatically classified** by using the trained classifier

Shallow Machine Learning for Pixel Classification

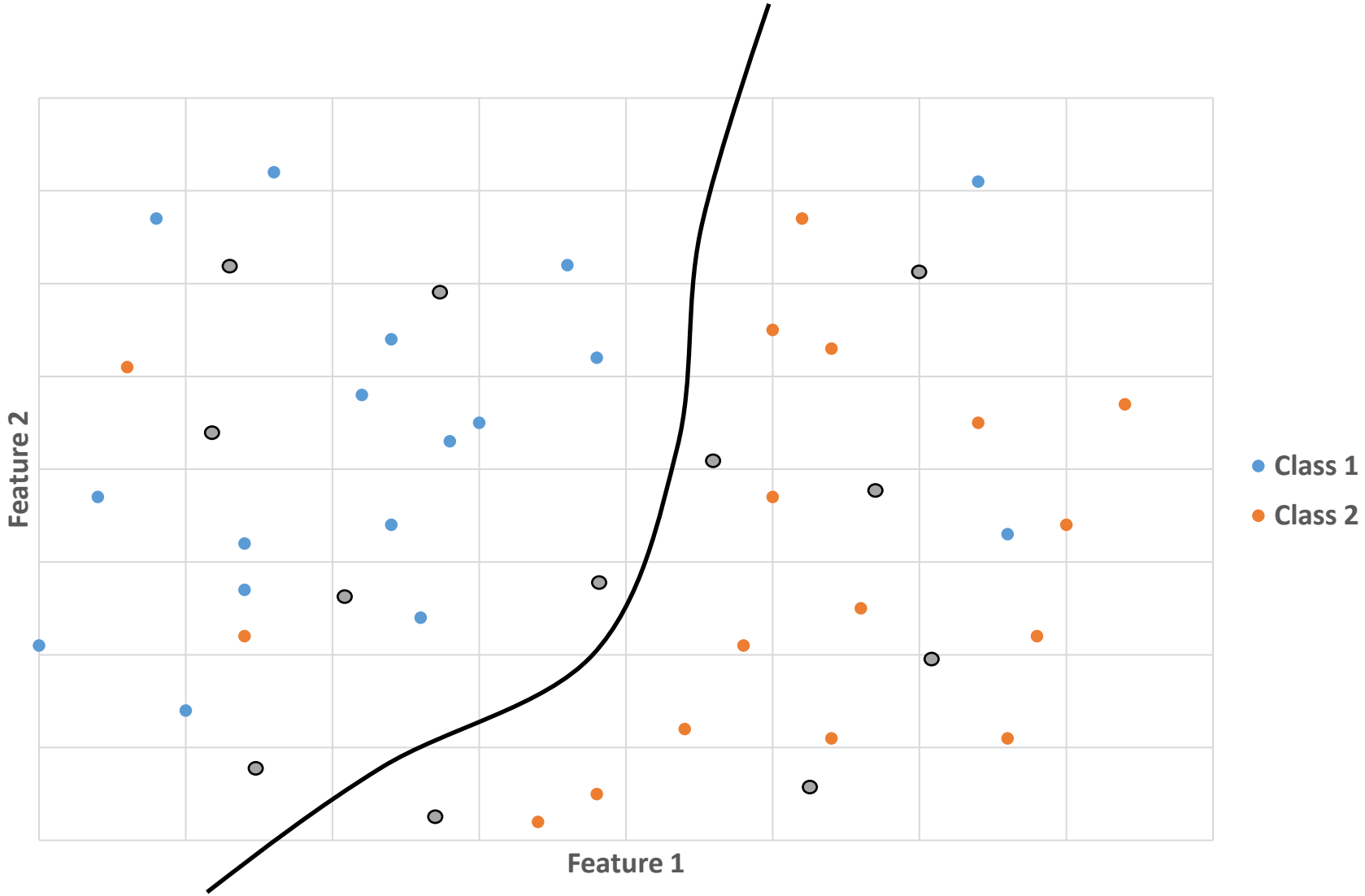


Shallow Machine Learning for Pixel Classification



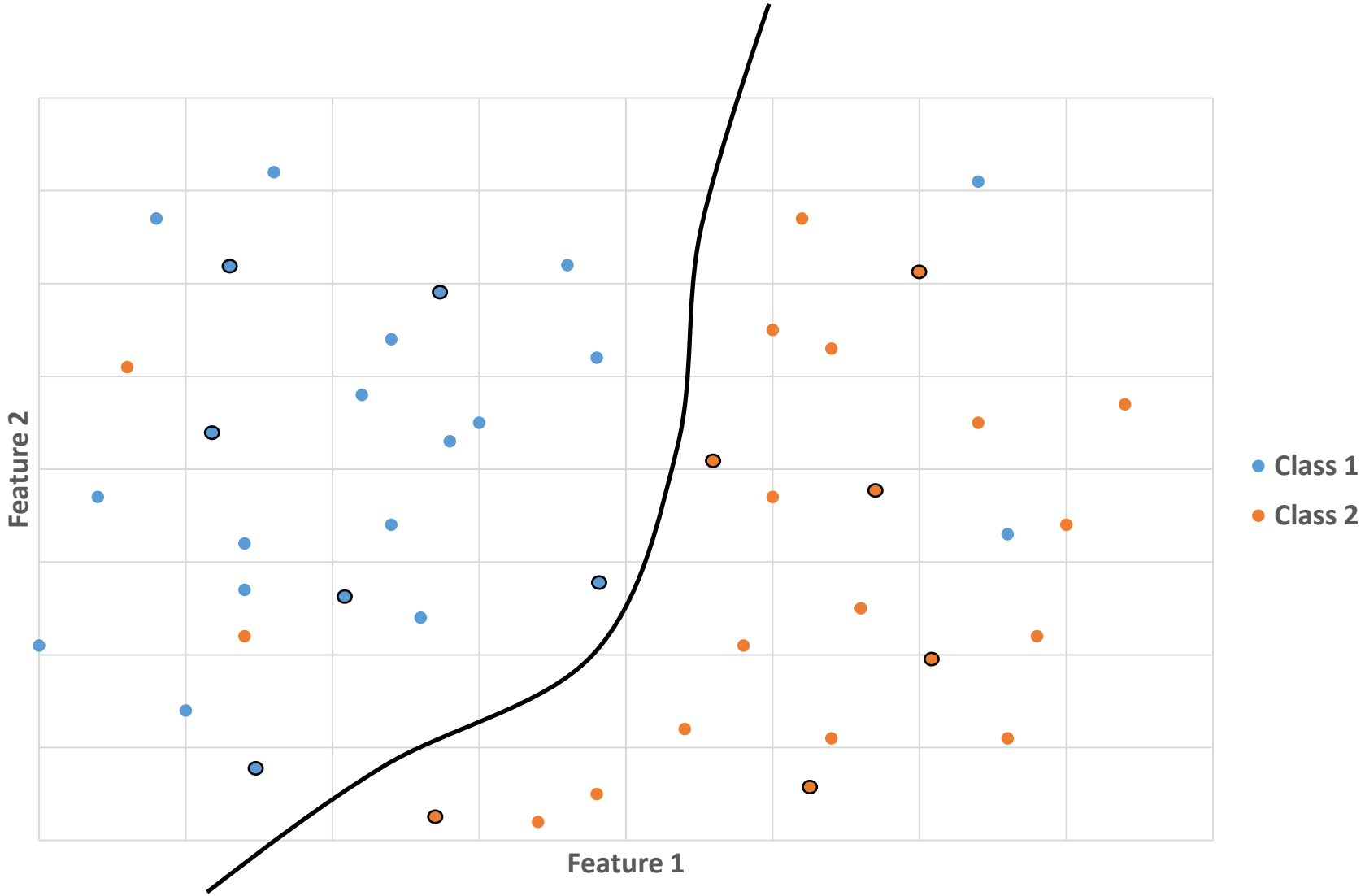
Training a classifier consists in **estimating** the **"best" separation** between classes

Shallow Machine Learning for Pixel Classification



The **estimated class** for new data will be given by the **trained classifier**

Shallow Machine Learning for Pixel Classification



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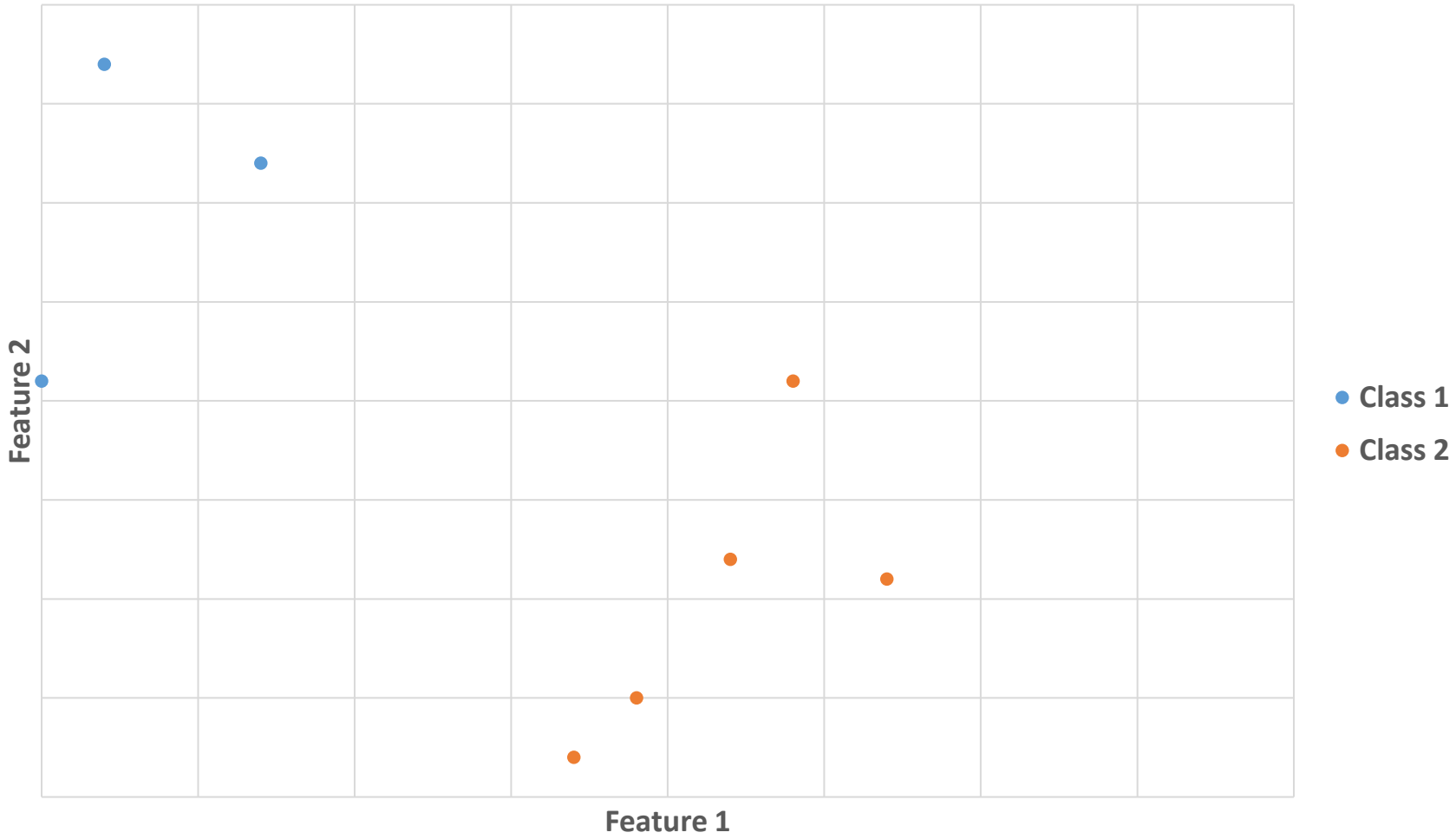
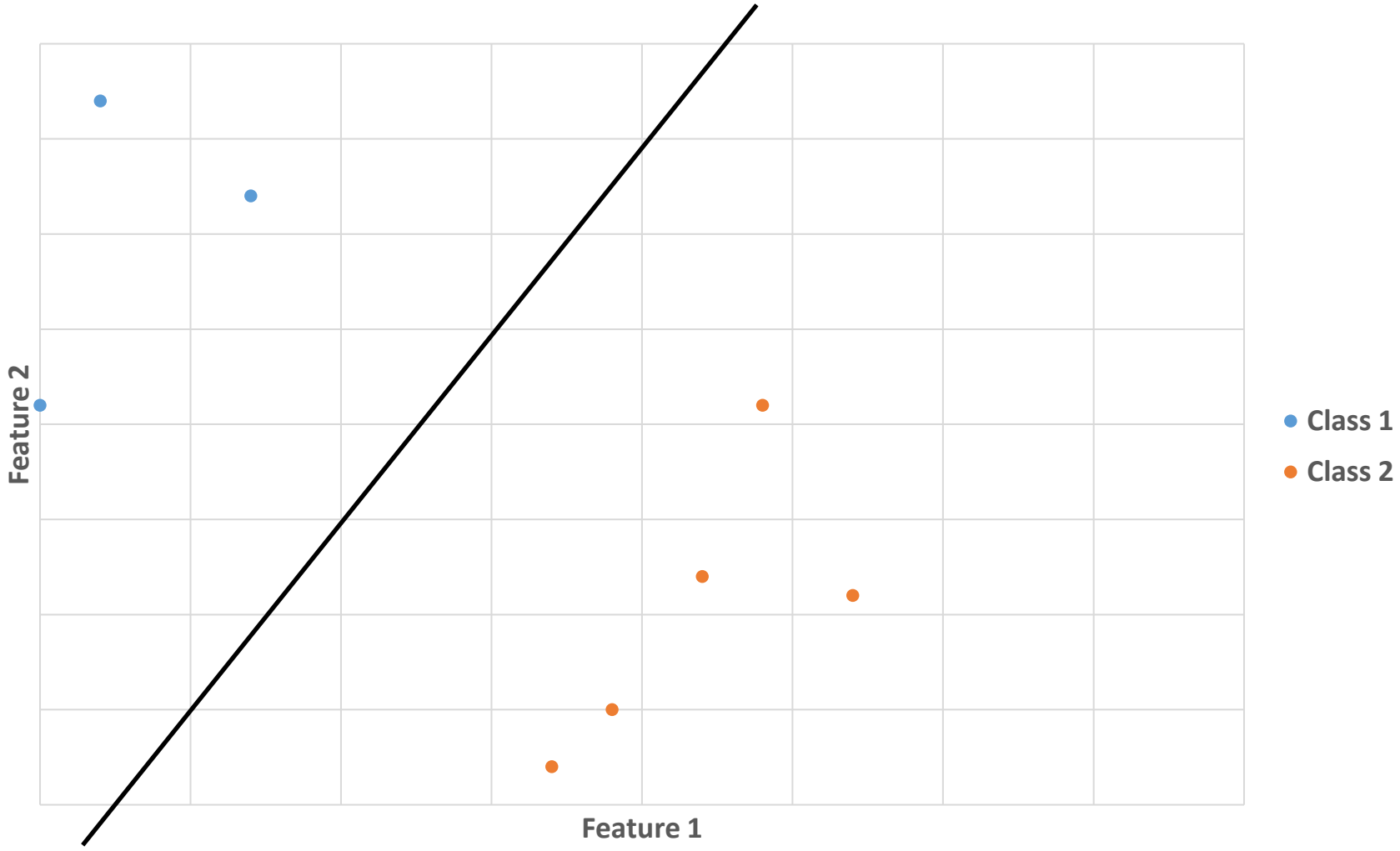


Training will be easier if **enough** and **representative** data is used

Shallow Machine Learning for Pixel Classification



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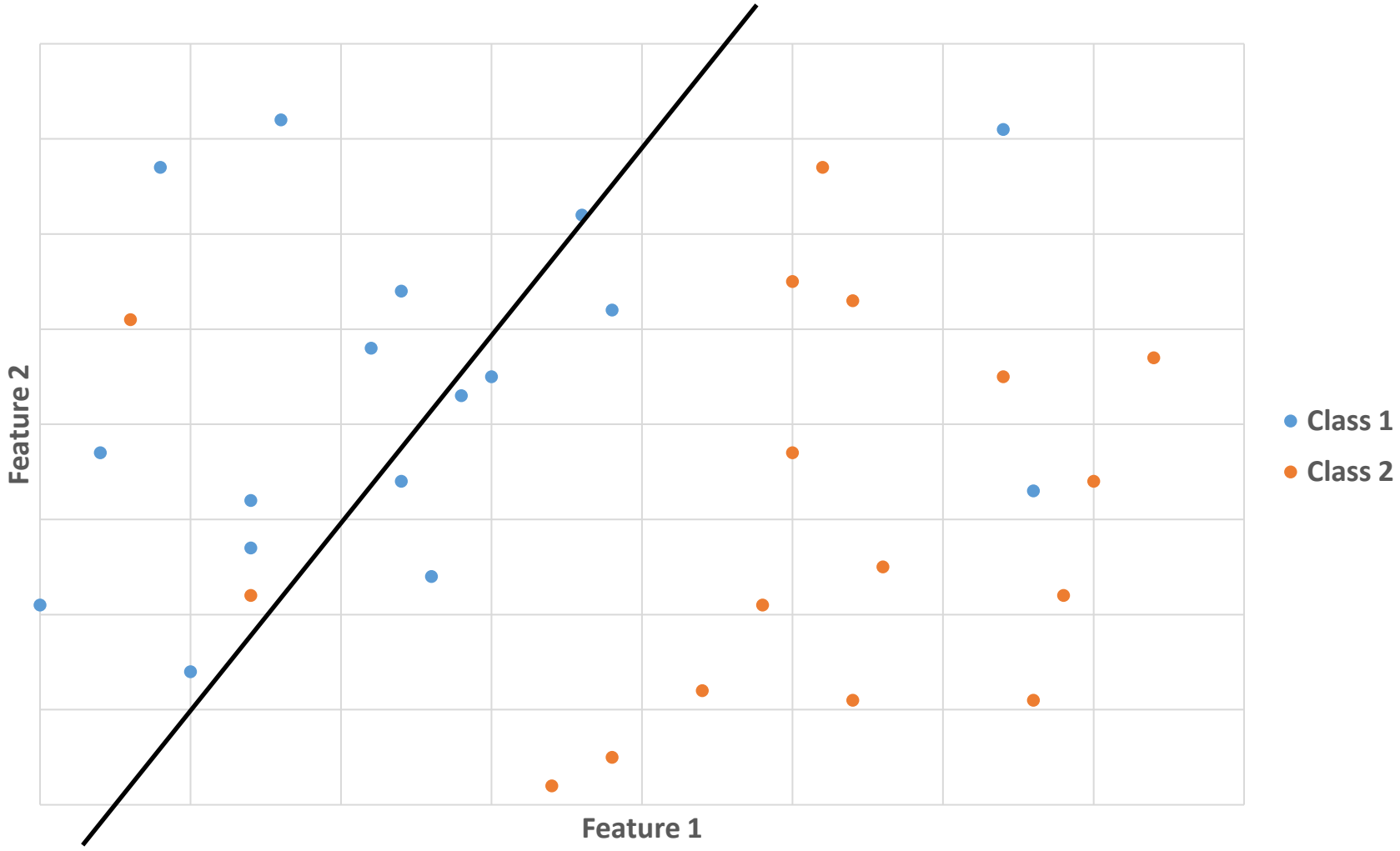

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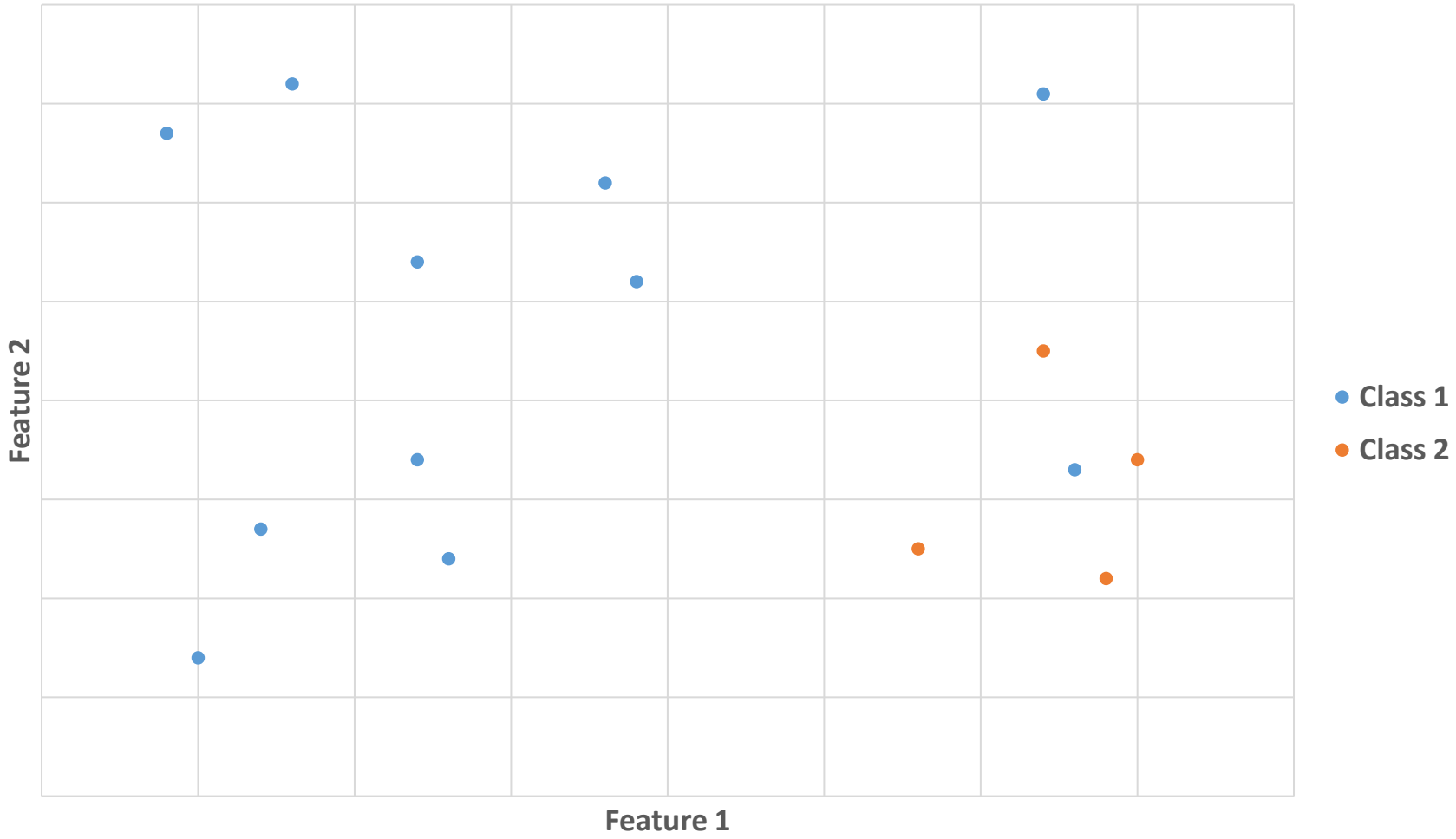
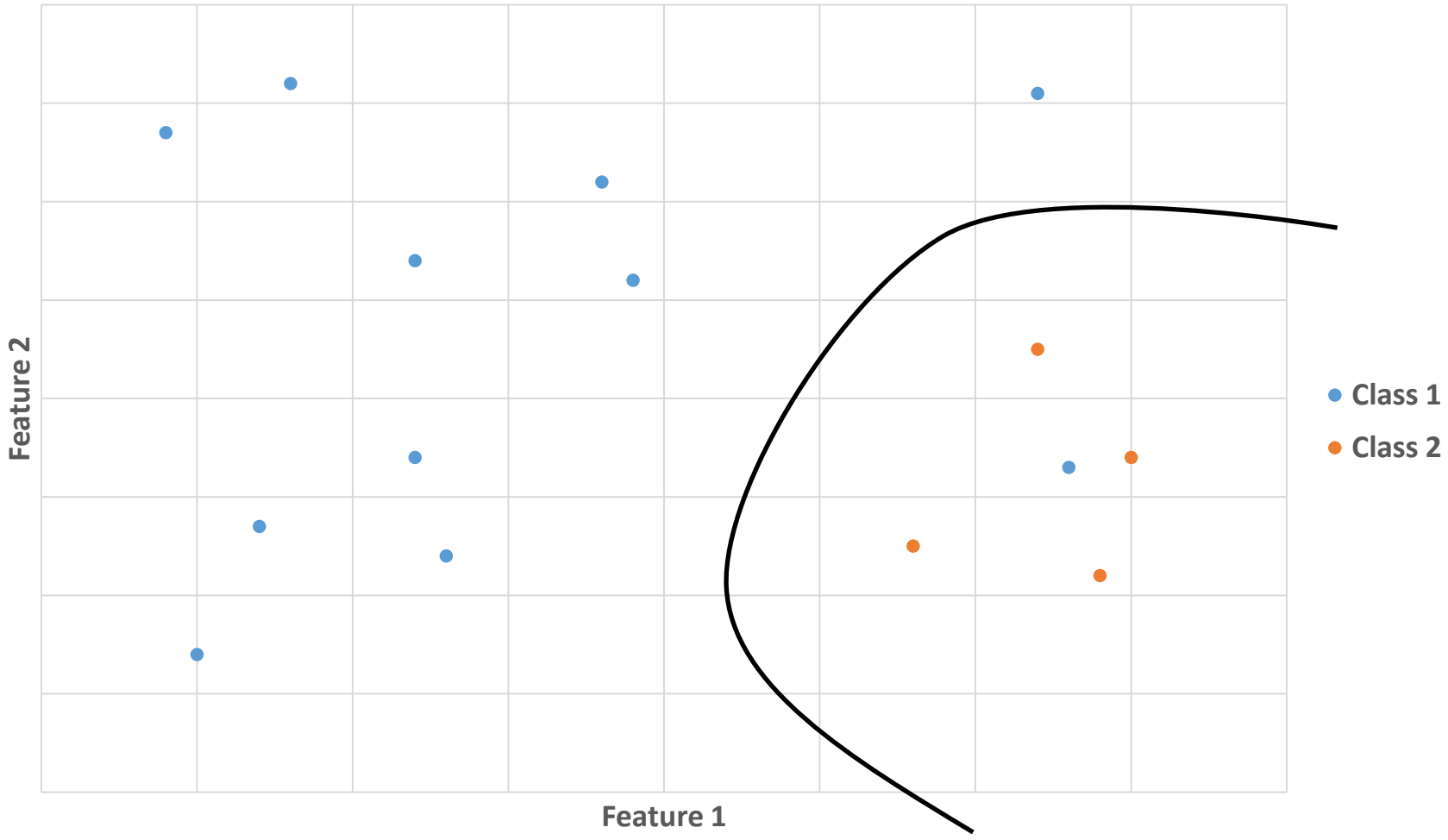


Class imbalance makes the training **more difficult**

Shallow Machine Learning for Pixel Classification



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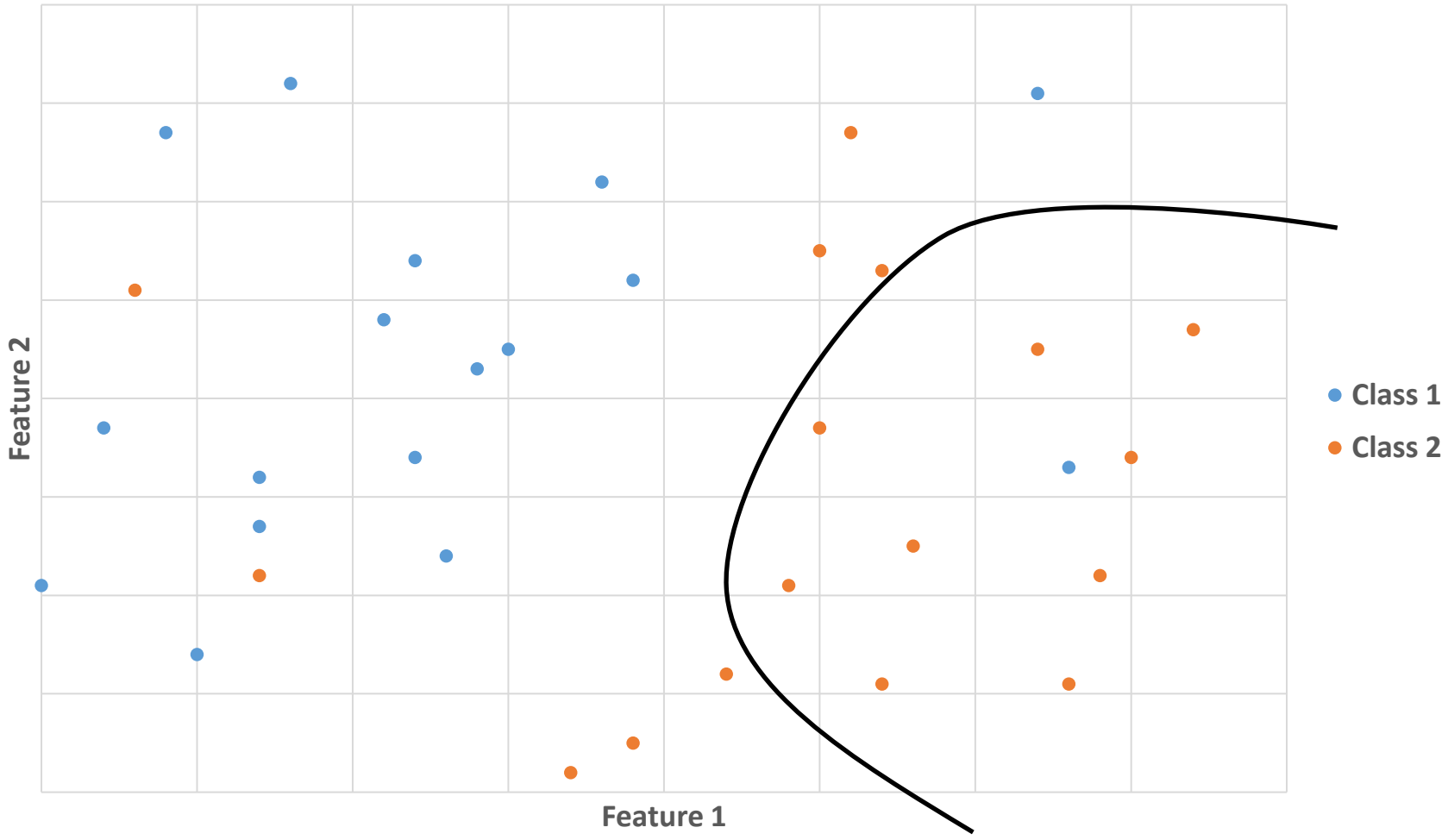
Shallow Machine Learning for Pixel Classification



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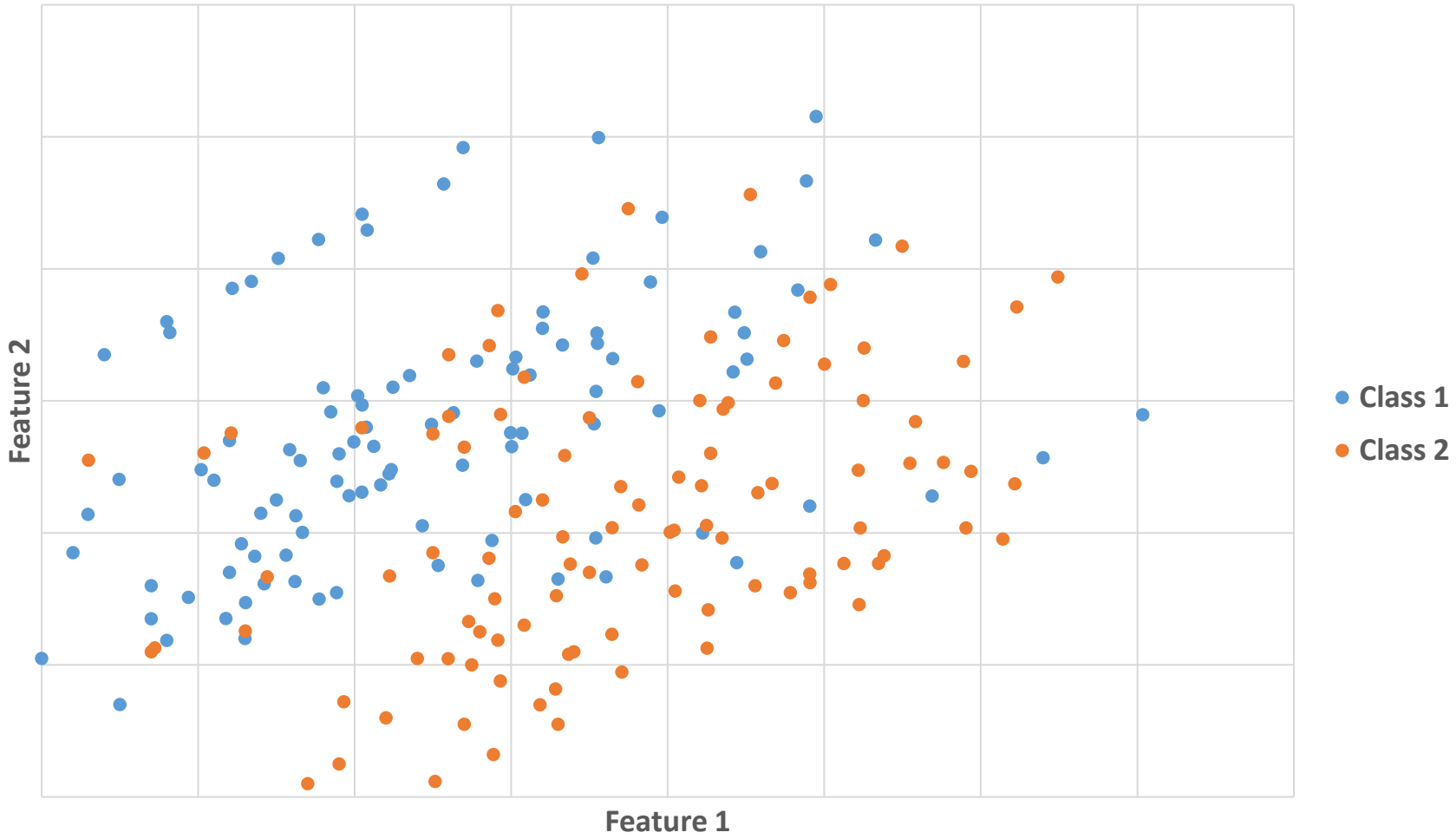
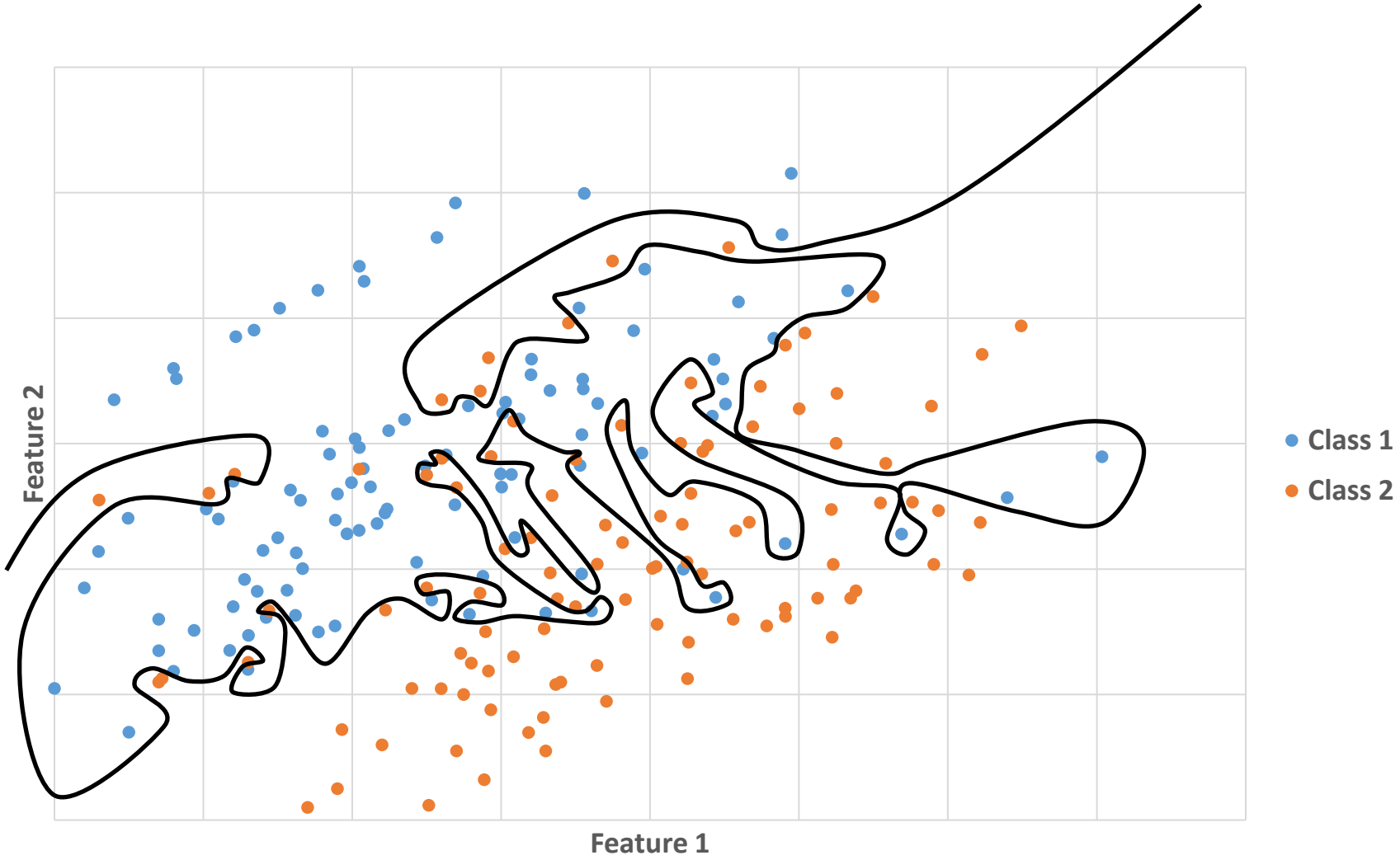
Shallow Machine Learning for Pixel Classification



Too many annotations can lead to **over-fitting**: works great on training data but poorly on new data

Shallow Machine Learning for Pixel Classification



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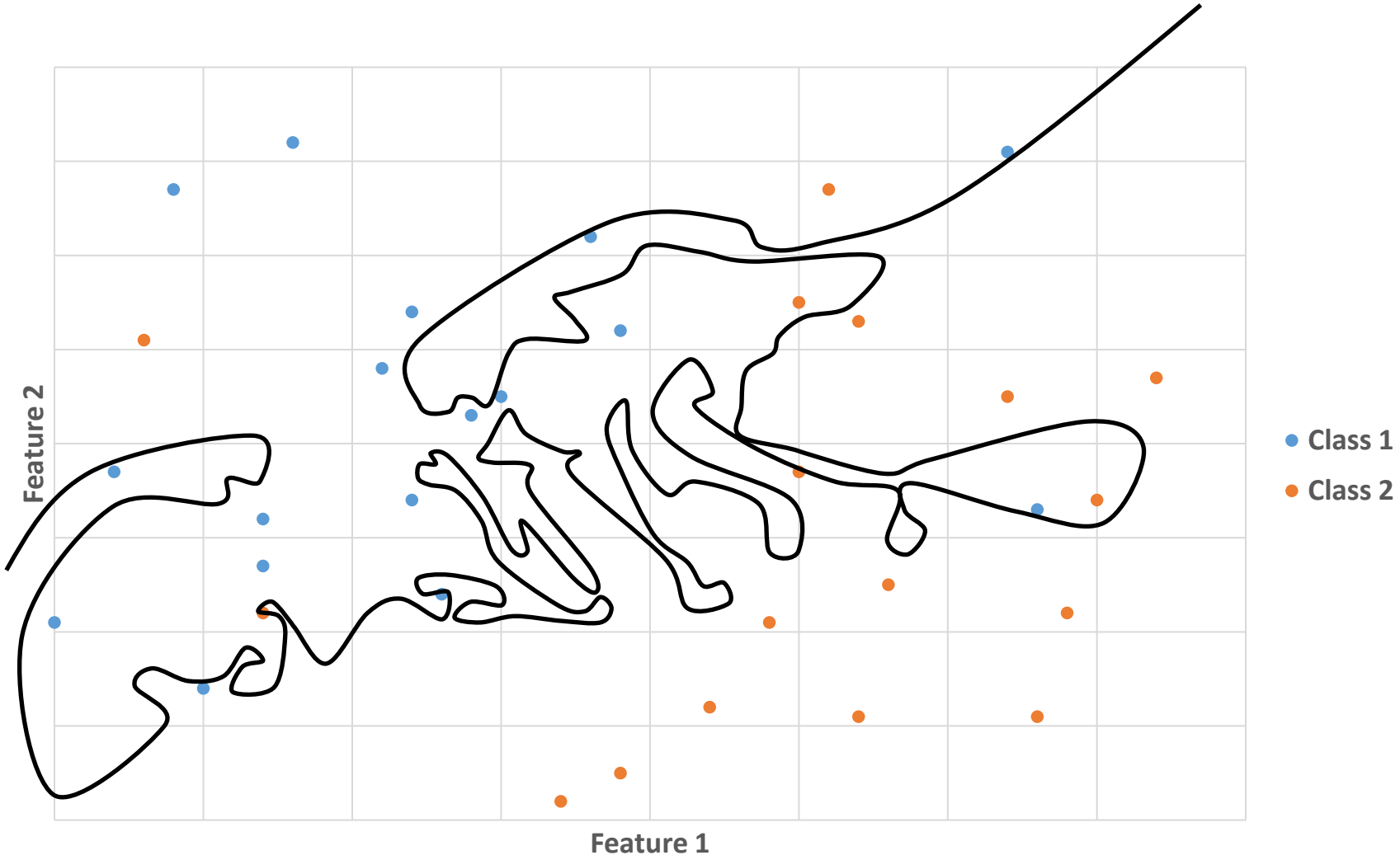
Shallow Machine Learning for Pixel Classification



Too many annotations can lead to **over-fitting**: works great on training data but poorly on new data

Shallow Machine Learning for Pixel Classification

- Select **image features appropriate** to the classification problem
- Manually annotate regions/objects that are **representative** of what is seen in images
- Use **small lines** for annotations to **avoid over-representation**
- Define roughly the **same amount** of annotations for **each class**
- **Do not** manually annotate an **entire region of slide** to avoid over-fitting

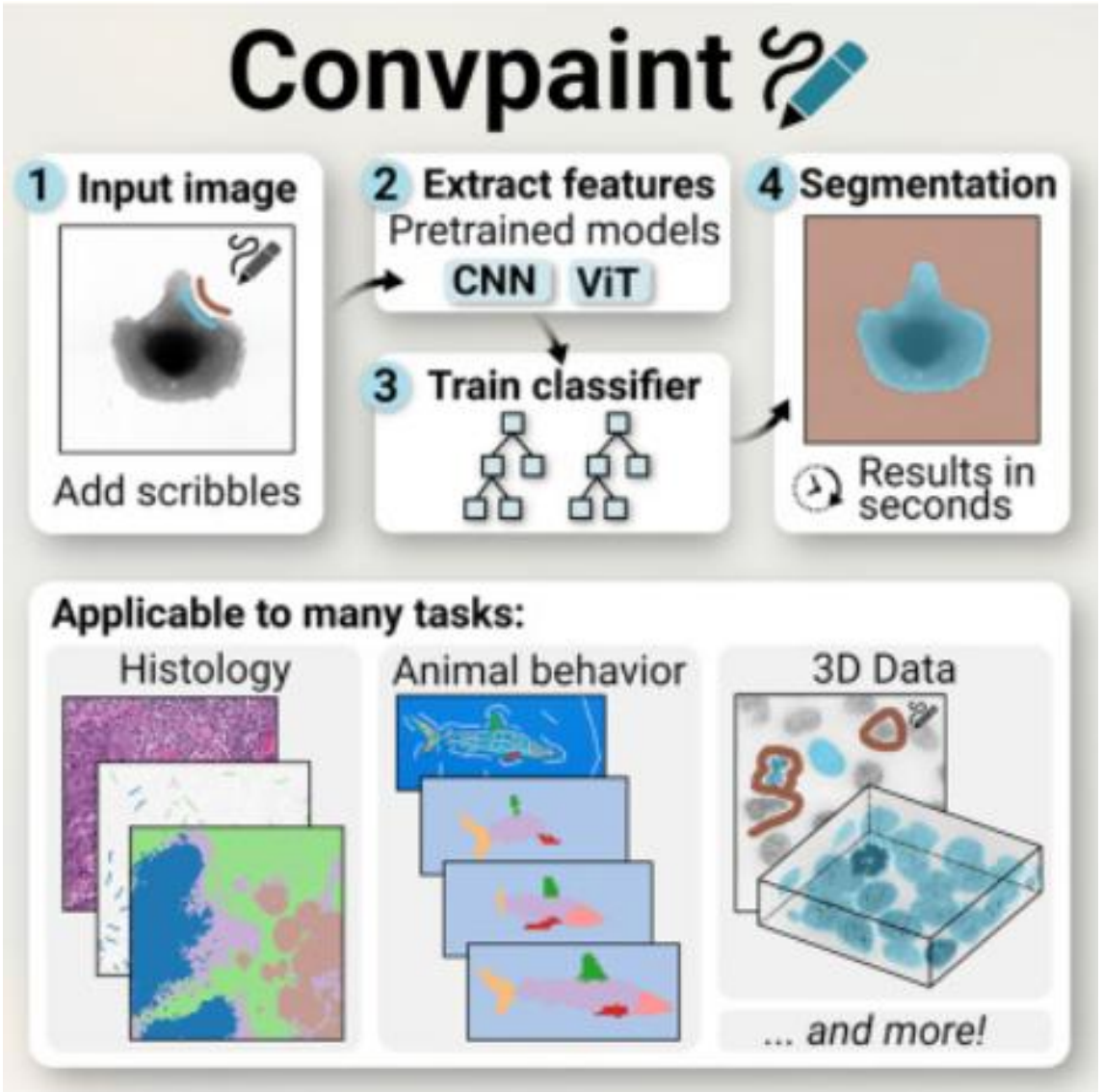
Cell Reports Methods

Volume 6, Issue 3, 23 March 2026, 101335

Article

Convpaint—Interactive pixel classification using pretrained neural networks

Lucien Hinderling [1 2], Roman Schwob [3], Guillaume Witz [3], Ana Stojiljković [3], Maciej Dobrzyński [1], Mykhailo Vladymyrov [3], Joël Frei [1], Benjamin Grädel [1 2], Agne Frismantiene [1], Olivier Pertz [1 4]



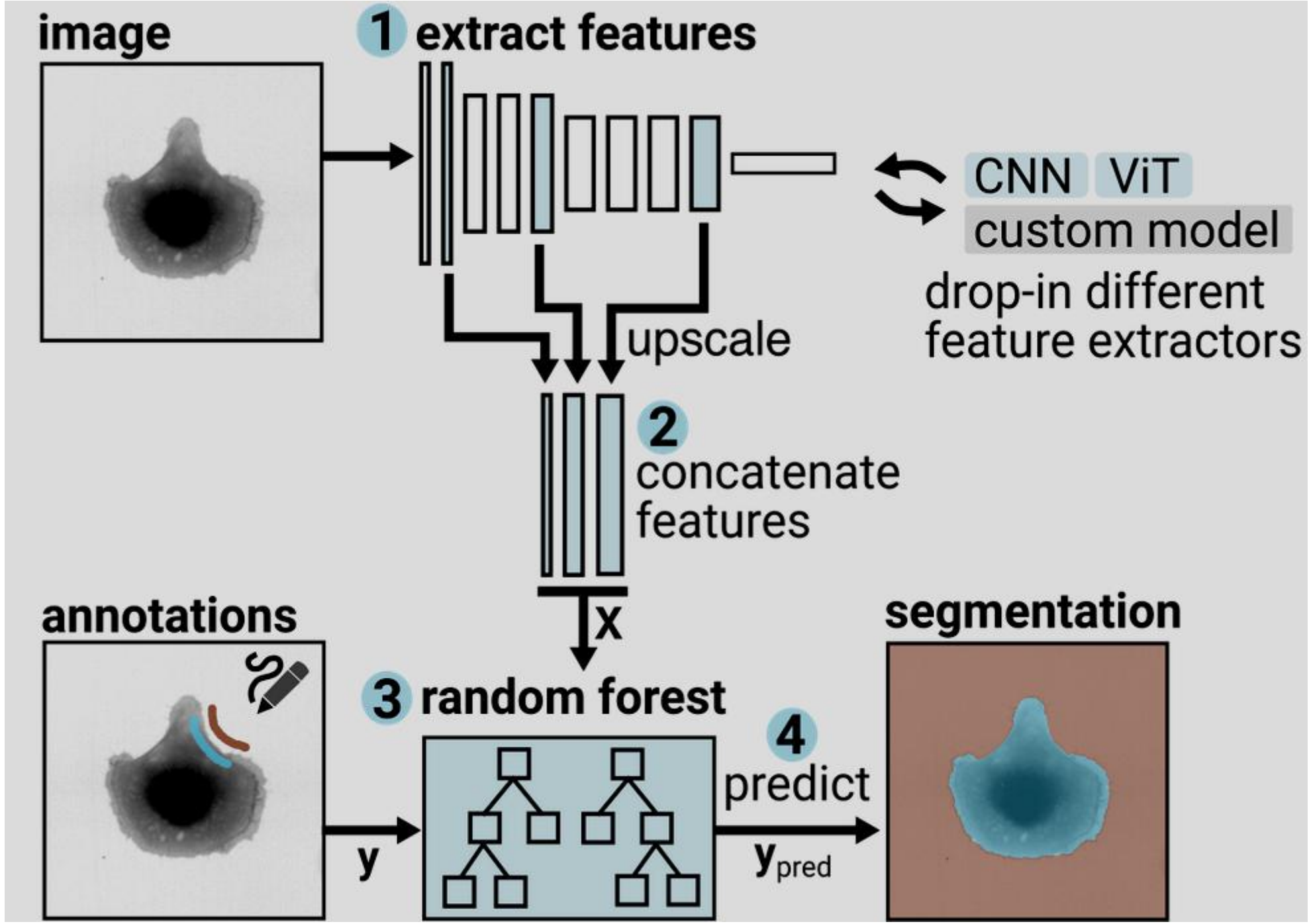


image
1 extract features
CNN
ViT
custom model
drop-in different
feature extractors
upscale
2
concatenate
features
x
annotations
segmentation
3 random forest
4
predict
y
y pred

- Open MutX_DR5v2_6_4_M1_cells.tif and MutX_DR5v2_6_4_M1_nuclei.tif (available at https://osf.io/uzq3w/), change colors, visualize both channels
- Use ConvPaint to segment nuclei
- Use ConvPaint to segment cells